Multi-Omics Target Discovery Package: Pancreatic Ductal Adenocarcinoma

BIOT 6900 · Portfolio Deliverable 1 (Part I synthesis, Labs 2–4) · Mohammed Sharukh · October 29, 2026
Repository: github.com/MohammedSharukh7/biot6900

1. Disease and question

Pancreatic ductal adenocarcinoma (PDAC) has one of the lowest five-year survival rates of any common solid tumor, and its four defining drivers are poor drug targets: *TP53*, *CDKN2A* and *SMAD4* are tumor suppressors inactivated by loss of function, and *KRAS* was considered undruggable until allele-specific inhibitors appeared (Cancer Genome Atlas Research Network, 2017). The tumor is also wrapped in a dense desmoplastic stroma that limits drug delivery. In this setting a useful target is one that (i) is increased at the **protein** level in tumor tissue, since protein is what a drug binds; (ii) is supported by more than one molecular layer, so the signal is unlikely to be noise; (iii) has a bindable site or an accessible extracellular surface matched to an existing modality; and (iv) has enough structural confidence to justify committing chemistry. The question for Part I was: starting from unbiased CPTAC-PDAC multi-omics, which genes meet all four criteria?

2. Multi-omics integration (Lab 2)

CPTAC-PDAC data were obtained from LinkedOmics (Cao et al., 2021; Vasaikar et al., 2018): RNA-seq (140 tumors, 21 normals), gene-level proteomics (140 tumors, 75 normals) and somatic mutation calls (140 tumors). Each layer was reduced to a gene-level tumor-versus-normal statistic, joined on gene symbol, and combined into an equal-weight rank-percentile score. **2,532** genes survived the three-layer join. RNA and protein changed in the same direction for **65.2%** of them (1,650 concordant, 882 discordant; Figure 1A). Concordance rose to 91.8% among the 170 genes significant in both layers, and to 14 of 15 in the top 15. Agreement is weak where effects are small and strong where they are large, as expected when noise dominates small fold-changes. Figure 2 shows how genes passing each layer's evidence threshold overlap. The layers are very unequal (RNA 311, protein 603, mutation 29), and most expression-supported genes (164) are shared by RNA and protein only.

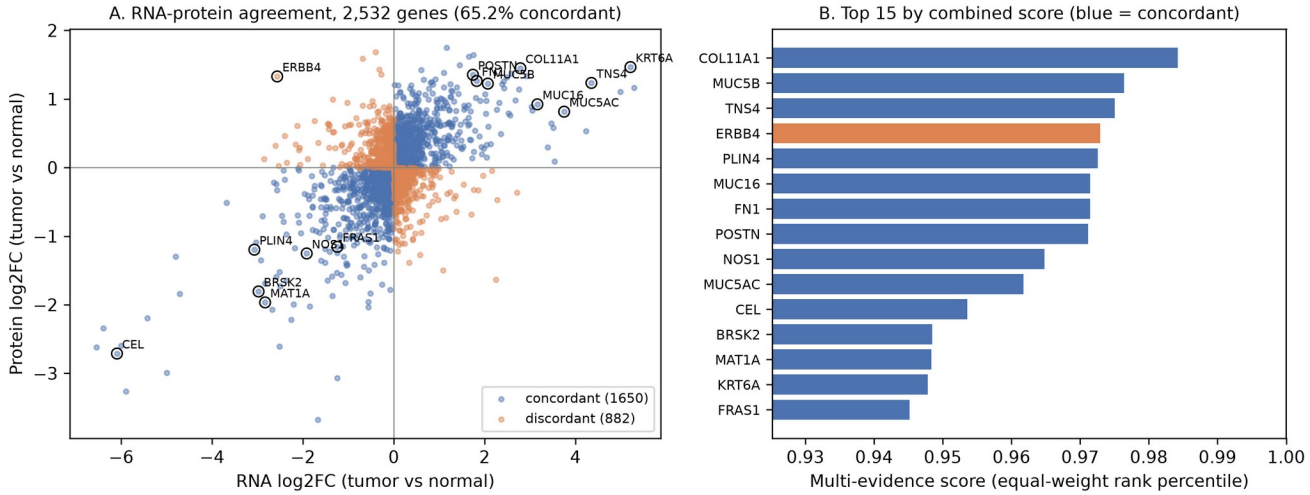


Figure 1. Multi-omics integration. (A) RNA versus protein log2 fold-change for all genes surviving the join; concordant genes in blue, discordant in orange, top-15 genes circled. (B) The top 15 genes by combined score. Data: CPTAC-PDAC via LinkedOmics.

Known-gene recovery. The mutation layer recovered the canonical PDAC drivers at the expected rank order: KRAS 96.4% (−log10 p = 260.8), TP53 75.0% (176.6), CDKN2A 20.7% (28.4) and SMAD4 17.9% (22.9), matching TCGA and CPTAC (Cancer Genome Atlas Research Network, 2017; Cao et al., 2021). CDKN2A and SMAD4 sit below their published alteration rates because both are frequently lost by homozygous deletion, which point-mutation calls do not capture. This is a sanity check that the data and pipeline behave correctly.

None of the four drivers entered the top 15 by combined score. This is not a failure: drivers act by changing protein *function* rather than abundance, while the score rewards large tumor-versus-normal swings in RNA and protein. The top of the list (Table 1) is therefore dominated by markers of PDAC tissue state, namely the desmoplastic stroma (COL11A1, FN1, POSTN) and the mucinous epithelium (MUC5B, MUC5AC, MUC16). In practice the genomic layer contributed little to the top 10: eight of the ten have mutation −log10 p ≤ 1.26 (p ≥ 0.05). The two exceptions, MUC16 (6.64) and MUC5B (2.50), are among the

longest genes in the genome, and long genes accumulate passenger mutations simply through their size. Their mutation signal should be read as a gene-length artifact, not as evidence of selection. The Venn diagram makes the point sharply: only six genes are supported by all three layers (CDKN2A, LRP1B, MUC16, MUC4, MUC5AC, MUC5B). Apart from CDKN2A, all of them are very long genes (LRP1B spans about 1.9 Mb), so "supported by all three layers" is not, on its own, a mark of a good target.

Gene	RNA log2FC	Protein log2FC	Mutation −log10 p	RNA–protein	Combined score
COL11A1	+2.787	+1.448	1.263	concordant	0.984
MUC5B	+2.071	+1.224	2.499	concordant	0.976
TNS4	+4.351	+1.234	0.771	concordant	0.975
ERBB4	**−2.565**	**+1.329**	0.771	**discordant**	0.973
PLIN4	−3.068	−1.197	0.771	concordant	0.973
MUC16	+3.162	+0.922	6.636	concordant	0.971
FN1	+1.826	+1.263	1.263	concordant	0.971
POSTN	+1.743	+1.353	1.263	concordant	0.971
NOS1	−1.916	−1.252	0.771	concordant	0.965
MUC5AC	+3.751	+0.815	3.220	concordant	0.962

Table 1. Top 10 genes by multi-evidence score (Lab 2, targets_pdac.csv). Nine of ten are RNA–protein concordant; ERBB4 is the exception.

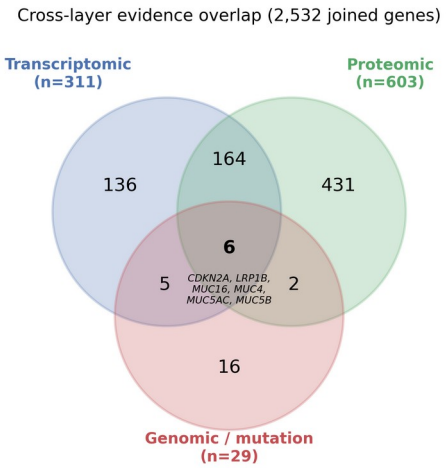


Figure 2. Genes passing each layer's evidence threshold (RNA |log2FC| ≥ 1 and p < 0.05; protein |log2FC| ≥ 0.5 and p < 0.05; mutation enrichment p < 0.05) among the 2,532 joined genes.

The discordant hit: ERBB4. ERBB4 is the only discordant gene in the top 10. Its mRNA falls in tumors (log2FC −2.565), consistent with an earlier report of reduced ERBB4 mRNA in pancreatic cancer, yet its protein rises (+1.329). The same study still detected ERBB4 protein in tumor cells in 61 of 75 cancers by immunostaining (Graber et al., 1999), so the protein signal is plausibly tumor-derived. Several explanations fit. The protein could be more stable in tumor cells. The proteomic peptides could come largely from the soluble intracellular domain (4ICD) that γ-secretase cleaves from the receptor, so the protein measurement may not report full-length, drug-relevant receptor. Or the signals could come from different cells, because a bulk sample mixes epithelium and stroma. Bulk, group-level data cannot tell these apart, and this matters because ERBB4 ranks first in Section 5.

3. Druggability triage (Lab 3)

All 15 Lab 2 targets mapped to the reviewed human proteome and were embedded with ESM C 300M (EvolutionaryScale, 2024). Small-molecule (p_{sm}) and antibody (p_{ab}) tractability priors were trained on about 1,200 proteins labeled by clinical precedence (Ochoa et al., 2023), after removing my own targets from the training panel to prevent leakage. Family-split validation gave AUROC 0.737 and AP 0.541, against 0.845 and 0.726 for a random split. That gap is the expected homology leakage: the prior is moderately informative, but part of its accuracy comes from recognizing relatives of already-drugged proteins. Targets were re-ranked as 60% Lab 2 evidence and 40% best-modality prior (targets_pdac_w3.csv).

Re-ranking moved **ERBB4** to 2nd (final 0.853; p_{sm} 0.702, p_{ab} 0.083). Its nearest drugged neighbors include the receptor tyrosine kinases EPHA3 and EPHA5 (cosine 0.95–0.96), so its prior is a kinase-family argument. **CEL** rose to 6th (0.600) on the highest p_{sm} of all 15 (0.717). Its visible nearest neighbors (MMP9, TNKS2; 0.91) are not close homologs, however, so its prior is less

firmly anchored than ERBB4's. **TNS4** stayed 3rd (p_{sm} 0.214). **POSTN** fell to 11th: its p_{ab} is only 0.048, so the prior does *not* support the antibody route its biology suggests. The new 1st place, **MUC5B** (0.880), and 7th, MUC5AC, were lifted by high antibody priors (p_{ab} 0.545 and 0.592).

Seven top-ranked genes were excluded from structure prediction for reasons the prior cannot see. MUC5B (5,762 residues), MUC5AC and MUC16 exceed the 2,700-residue AlphaFold Database limit and would be fragmented, and their mutation evidence is length-driven (Section 2). COL11A1 functions as a triple helix, which a single-chain model cannot represent. FN1 is a large array of repeated modules. PLIN4 and NOS1 are *decreased* in tumors, so inhibiting them has no therapeutic logic. ERBB4 and CEL went forward to Lab 4 as the two strongest small-molecule priors, to test whether a sequence-based prior predicts a real pocket.

4. Structural assessment (Lab 4)

ERBB4 (UniProt Q15303, 1,308 aa). The AlphaFold model (Jumper et al., 2021; Varadi et al., 2022) has mean pLDDT 73.1, with 43.8% of residues above 90 and 26.9% below 50 (Figure 3A). The low-confidence stretches are the signal peptide (1–24), the segments flanking the transmembrane helix (637–652 and 688–708), and the entire C-terminal tail (1025–1308). That tail is a disordered phosphotyrosine docking region, so no binding site can be claimed there. PAE resolves two confident domains, the ectodomain (27–635) and the kinase (711–1022), with a mean inter-domain error of 28.7 Å (Figure 3B). Their relative placement is therefore unknown, as expected for domains joined through a single membrane-spanning helix, and the kinase was analyzed on its own.

Superimposed on PDB 2R4B chain A (2.40 Å, kinase domain bound to a thienopyrimidine inhibitor covalently linked to Cys803; Wood et al., 2008) using residues with pLDDT ≥ 70, the kinase RMSD is 2.82 Å. That is above the ~2 Å typical of a good match, but the per-residue deviations show the excess comes from a few residues: the end of the crystallized construct (1022, 38.3 Å; 988, 10.6 Å) and a short loop at 765–768 (about 4–5 Å). The core of the kinase agrees to well under 2 Å. fpocket (Le Guilloux et al., 2009) ranked pocket 93 first: druggability score 0.894, volume 1,067 $Å^3$, lining mean pLDDT 80.9 and PAE 5.4 Å, verdict keep. This pocket covers 15 of the 22 residues that contact the crystallographic inhibitor (68%), including Cys803. The seven it misses (750, 781–783, 795, 862, 864) are at the cleft edges. Of the remaining top-five pockets, one was marked suspect (PAE 13.1 Å) and one rejected (pLDDT 54.4). The top-ranked pocket in a confidently predicted region is therefore the site where real ErbB inhibitors bind (Qiu et al., 2008; Wood et al., 2008). **Call: structure supports a small-molecule pocket.**

CEL (UniProt P19835, 753 aa). Mean pLDDT is 82.6 (66.3% above 90, 16.2% below 50). The low-confidence stretches are the signal peptide (3–17 of residues 1–20) and the C-terminal region (555–592 and 717–753, with residues in between only at pLDDT ~40–60). That region is the O-glycosylated proline-rich repeat region, disordered in solution, where frameshifts cause MODY8. In the 3D model (Figure 3D) AlphaFold packs these tandem repeats into a regular stacked array at pLDDT 50–70, an arrangement that should not be trusted. PAE gives one confident domain, residues 21–552, which is exactly the mature catalytic domain. Against 1F6W chain A (residues 1–538, 2.3 Å, no bound ligand; Terzyan et al., 2000) the RMSD is 1.35 Å. The largest deviations (444–449, up to 9.1 Å) lie in a loop just before catalytic His455. Smaller peaks near residues 140 and 297 match the two mobile substrate-binding loops the crystallographers reported (mature 115–125 and 270–285, about 20 residues later in UniProt numbering).

The pocket result is the surprise. The best kept pocket (56; score 0.925, pLDDT 93.7, PAE 2.1 Å) does **not** contain the catalytic triad (Ser214, Asp340, His455). Part of its lining (128–131, 287–301) sits on or beside those mobile loops. The pocket that does hold Ser214 and His455 (pocket 61) is confidently modeled (pLDDT 95.6) but scores only 0.135. A likely reading is that, in this static model without bile salt, the active site is partly closed by loops that move during activation, so fpocket scores a neighboring groove instead. With no ligand in 1F6W there is no experimental site to check against. **Call: possible but uncertain.** The Lab 3 prior was therefore confirmed outright for ERBB4 but only partly for CEL: CEL has a pocket, but not demonstrably at its functional site.

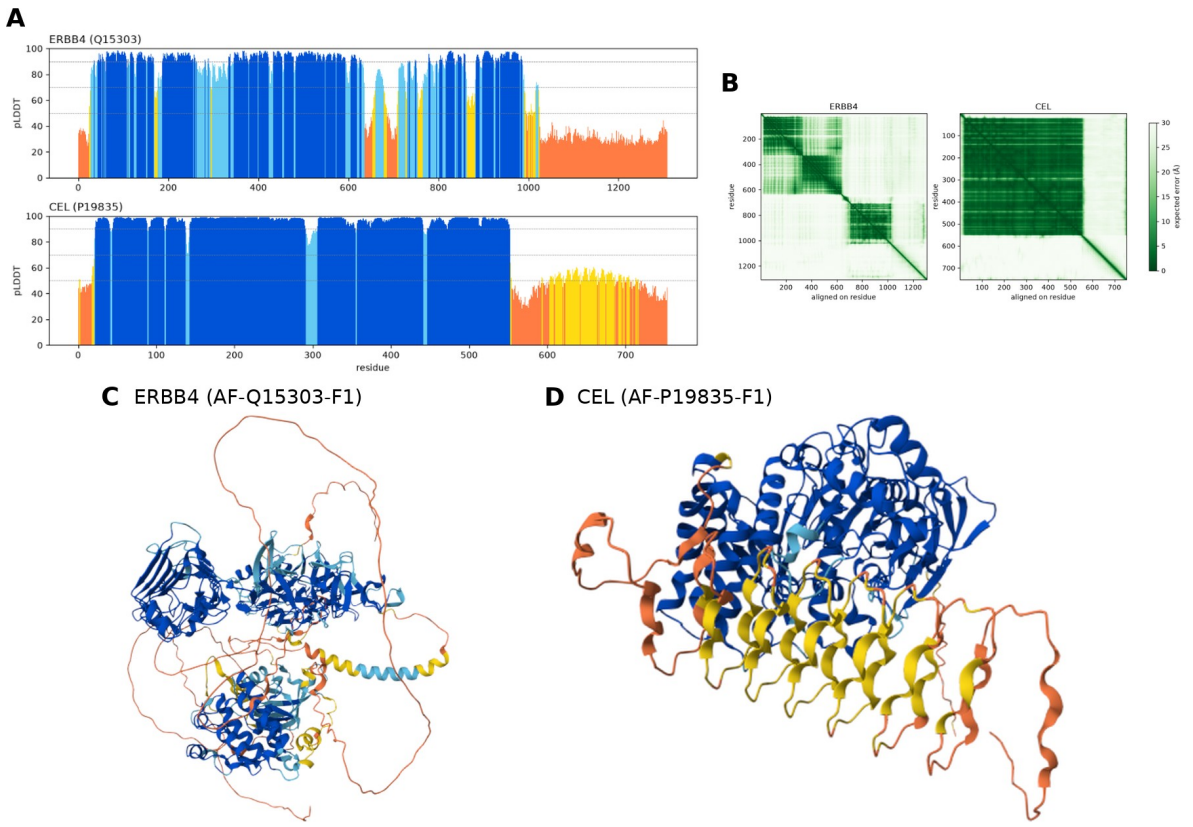


Figure 3. Lab 4 structure assessment of the AlphaFold models of ERBB4 (AF-Q15303-F1) and CEL (AF-P19835-F1). (A) Per-residue pLDDT: dark blue > 90, light blue 70–90, yellow 50–70, orange < 50. (B) Predicted aligned error; dark blocks are confident domains. (C, D) 3D models from the AlphaFold Database, colored by pLDDT on the same scale. ERBB4 shows its ectodomain and kinase as separate compact domains with no confident arrangement between them, and its C-terminal tail as disordered orange coil. CEL shows a compact catalytic domain with the C-terminal repeat region packed beneath it at low confidence.

Metric	ERBB4	CEL	Reading
Mean pLDDT / % < 50	73.1 / 26.9%	82.6 / 16.2%	Both cores confident; termini disordered
Confident domains (PAE)	2: 27–635, 711–1022; inter-domain 28.7 Å	1: 21–552	ERBB4 domains analyzed separately
RMSD vs PDB (pLDDT ≥ 70)	2.82 Å vs 2R4B A (terminal outliers)	1.35 Å vs 1F6W A	Prediction matches experiment
Best kept pocket	#93: 0.894, pLDDT 80.9, PAE 5.4 Å	#56: 0.925, pLDDT 93.7, PAE 2.1 Å	Both pass the confidence filters
Functional-site check	Covers 68% of inhibitor site, incl. Cys803	Misses triad; triad pocket #61 scores 0.135	ERBB4 validated; CEL ambiguous
Druggability call	**Supports a small-molecule pocket**	**Possible but uncertain**	

Table 2. Structural quality metrics and pocket results (structure_assessment_pdac.csv).

5. Prioritized target package

Table 3 integrates all three layers. The ranking uses the four criteria from Section 1, in order: tumor-increased protein, multi-layer support, modality fit, and structural confidence.

#	Target	Multi-omics (Lab 2)	Tractability (Lab 3)	Structure (Lab 4)	Decision and rationale
1	**ERBB4**	Rank 4 (0.973); protein +1.33 but RNA −2.57 (discordant)	Re-rank 2; p_sm 0.70, p_ab 0.08	Kinase 2.82 Å vs 2R4B; pocket 0.894 covers inhibitor site	Advance: small molecule. Clears every layer; covalent pan-HER inhibitors (afatinib, neratinib, dacomitinib) enable repurposing. Gate: confirm full-length receptor in tumor cells.
2	**TNS4**	Rank 3; largest RNA rise (+4.35), protein +1.23, concordant	Re-rank 3; p_sm 0.21	Not modeled (gap)	Advance to structure. Strongest, cleanest tumor-epithelial signal, ranked high in two layers, but an SH2/PTB scaffold with a modest prior. Next Lab 4

#	Target	Multi-omics (Lab 2)	Tractability (Lab 3)	Structure (Lab 4)	Decision and rationale
					target.
3	**POSTN**	Rank 8; concordant up (+1.74 / +1.35)	Re-rank 11; p_ab 0.05	Not modeled (gap)	Hold. Secreted stromal protein, so antibody-accessible in principle, but the prior disagrees and there is no structure yet.
4	**CEL**	Rank 11; strongly down (RNA −6.09, protein −2.71)	Re-rank 6; p_sm 0.72	1.35 Å vs 1F6W; best pocket misses triad	Deprioritize; keep as a pipeline control. Lost in tumors, and its top pocket is not the catalytic site.
—	MUC5B, MUC5AC, MUC16, COL11A1, FN1, PLIN4, NOS1	Top 10	MUC5B re-rank 1 and MUC5AC 7 (high p_ab)	Not modelable	Dropped from this pipeline: length, fold, or decreased direction. Mucins remain possible antibody/ADC targets outside it.

Table 3. Final ranked shortlist integrating Labs 2–4.

Why this order. ERBB4 ranks first because it is the only candidate that clears every layer. It has a top-5 multi-omics score driven by increased protein, the strongest kinase-family small-molecule prior, and a confidently predicted ATP pocket that recovers the inhibitor site of an experimental structure. Its weakness is the RNA/protein discordance and ERBB4's mixed literature, in which it has been reported as both oncogenic and growth-suppressive, so it advances with a validation gate rather than unconditionally. TNS4 ranks second because it scores highly in both Lab 2 and Lab 3 and has the largest concordant tumor increase. It lacks structural evidence and a strong prior, so its next step is structure prediction, not chemistry. POSTN has clean, desmoplasia-consistent biology, but two of the three layers do not yet support it: the antibody prior is low and no structure was assessed. CEL is the instructive case. It has the highest small-molecule prior and an excellent experimental-quality model, yet it fails the first criterion outright: it is one of the most strongly lost proteins in PDAC and ranks 11th only because the score rewards large changes in either direction. Its best pocket is not its catalytic site. A good pocket on the wrong protein is not a target. Finally, MUC5B tops the Lab 3 re-ranking, but it cannot be assessed structurally and its genomic support is a length artifact. A ranking that kept it first would be rewarding the score, not the biology.

6. Limitations and validation

What this analysis cannot claim. (1) Integration is group-level, not patient-paired, and normal sample counts differ by layer (21 RNA versus 75 protein). No statement about individual patients, or about within-patient RNA–protein coupling, is supported. (2) Bulk tissue mixes cell types. Acinar loss and stromal expansion in tumors push acinar genes (CEL) down and stromal genes (COL11A1, POSTN) up, so part of the score measures tissue composition rather than tumor-cell biology. (3) The mutation layer misses deletions and copy-number changes and is confounded by gene length. (4) Tractability priors are homology-inflated (family-split AUROC 0.737). (5) AlphaFold models are static, ligand-free monomers without glycans or membrane; pockets in regions with pLDDT < 70 were rejected, and conformational pockets (for example DFG-out) cannot be seen. (6) Every result is correlational. None shows that a gene drives PDAC.

Validating ERBB4. Step 1, localization: immunohistochemistry on a PDAC tissue microarray, using antibodies against both the ectodomain and the C-terminus, to establish whether the increased protein is full-length receptor in tumor epithelium or cleaved 4ICD or stromal signal. Paired per-patient CPTAC RNA and protein, plus isoform-level RNA-seq, would test the stability explanation. Step 2, dependency: ERBB4 knockdown or CRISPR knockout in PDAC cell lines and patient-derived organoids, measuring viability and phospho-ERBB4 and downstream pAKT/pERK. Step 3, pharmacology: dose–response to afatinib and neratinib in ERBB4-high versus ERBB4-low models, with target engagement confirmed by cellular thermal shift (CETSA). Selectivity against EGFR and HER2 would be the key go/no-go criterion.

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